

Quad22

A four-band compander, saturator and EQ channel strip.



Overview

Quad22 is a modern take on the classic studio noise-reduction compander idea, turned into a four-band channel strip. It splits the audio at 80 Hz, 3 kHz and 9 kHz with phase-coherent crossovers, so the bands add back together cleanly.

Each band runs through saturation, a fixed-frequency EQ, a one-knob compander and a gain control. The compander is the heart of it: turned left it gently lifts the quiet detail and tails in that band while leaving the peaks alone (the encode side); turned right it pushes the quiet parts further down (the decode side).

The bands are summed, blended with a phase-aligned dry signal by MIX, then go through the master TRIM and a soft ceiling limiter.

Reach for it for

- Bringing up air, breath and detail without boosting peaks
- Tightening a woolly low end or quietening noise in one band
- Gentle multiband glue and colour on a mix or bus
- Tape, transformer or transistor colour on just part of the spectrum

Controls

BAND 1 to BAND 4

Four bands with the same controls: BAND 1 below 80 Hz (SUB), BAND 2 from 80 Hz to 3 kHz (BODY), BAND 3 from 3 to 9 kHz (PRESENCE) and BAND 4 above 9 kHz (BREATH).

OFF / IRON / TAPE / XSTR	The saturation type. IRON (transformer): a symmetric soft clip, mostly 3rd harmonic. TAPE: a smooth, symmetrical saturation. XSTR (transistor): a lopsided curve that adds even harmonics. OFF: no saturation.
DRIVE	How hard the band hits the saturation, 0 to 100%. The output is partly level-compensated. To keep the lows clean, IRON and XSTR are driven less hard in BAND 1 (40%) and BAND 2 (75%).
LF SAFE (BAND 1)	Lights while IRON or XSTR is driving BAND 1, to show the low-frequency safeguard is in use.
LO SHELF / BELL / HI SHELF	The band's EQ, -12 to +12 dB, at a fixed frequency: BAND 1 a low shelf at 50 Hz, BAND 2 a bell at 800 Hz, BAND 3 a bell at 5 kHz, BAND 4 a high shelf at 12 kHz. The bells are broad.
COMPAND	One knob, -100 to +100, with a centre detent (FLAT). Left (LIFT): upward compression - material below about -20 dB is lifted, by up to 15 dB, while the peaks are left alone; the fast 3 ms attack lets transients through unlifted, so the tails and low-level detail come up. Right (EXP): downward expansion - the threshold rises from -50 to -26 dB and the ratio goes up to 1:2, cutting quiet parts by up to 40 dB. It never adds gain, so it cannot push the peaks over. BAND 1 recovers more slowly (200 ms rather than 75 ms) so the sub-bass doesn't flutter.
GAIN	The band's output level, -24 to +24 dB.
Meter	The band's output level, -48 to +3 dB.

SOLO	Solo in place; several bands can be soloed at once.
BYPASS	Passes the band through untouched: no saturation, EQ, compander or gain.

MASTER

MIX	Dry / wet balance, 0 to 100% (100% by default). The dry signal is phase-aligned with the bands, so partial settings blend cleanly.
TRIM	Master output level, -24 to +24 dB, before the ceiling.
CEILING	A soft limiter on the output, -12 to 0 dB (-0.3 dB by default): the peaks are rounded off smoothly towards this level.
OUT	The output level, -48 to +3 dB.

The header

Display	Shows the current preset, or the control you are touching and its value. Click it for the preset list.
< / > (PRESET)	Step through the presets.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the brushed-aluminium rack and the flat, soft panel.
Colour key (flat panel only)	Switches the flat panel between light grey and dark blue.

Drag the knobs up and down (hold Shift for fine moves); double-click or Ctrl-click to reset one.

Presets

Init	Everything flat: no saturation, EQ, compander or gain.
Air Lift	Lift on BAND 3 (-35) and BAND 4 (-50), with +1 and +2 dB of EQ.
Gentle Glue	Mild lift on all four bands, with light tape on BAND 2.
Tape Master	Low-drive tape on every band, +1 dB on the 50 Hz shelf and a touch less top.
Vocal Presence	+2 dB at 5 kHz with lift on BAND 3, BAND 2 slightly trimmed, +1 dB of air.
Low-End Control	Expansion on BAND 1 with a light transformer, the 800 Hz bell down 1 dB.
Analog Sheen	Transistor drive on BAND 4 with +2.5 dB of high shelf and lift.
Loudness Push	Lift on all bands, light transformer drive on BAND 1 and tape on BAND 2, master TRIM +2 dB.

Tips

- The presets set the saturation, drive, EQ, compander and gain of each band plus TRIM, CEILING and MIX; SOLO and BYPASS are left as they are.
- Start with COMPAND turned left on BAND 3 and BAND 4 to bring up breath and air on a vocal or mix without making it harsher - the peaks are left alone.
- Turn COMPAND right on BAND 1 to tighten a boomy low end, or on BAND 4 to push hiss and noise down in the quiet passages.
- SOLO a band while you set its compander: you hear exactly what it is lifting or pushing down.

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